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Submicron magnetite-enhanced tribromophenol removal and methanogenesis under microaerobic condition

Fan Yang, Jing Wang,* Chen Gu, Ying Han, Guang-fei Liu and Hong Lu

Abstract

BACKGROUND: To accelerate direct interspecies electron transfer (DIET), magnetite is supplemented in anaerobic activated sludge (AS). However, it is usually reduced by dissimilatory iron reducing bacteria (DIRB). Therefore, an AS system supplemented with submicron magnetite particles (Fe_3O_4/AS) in an up-flow microaerobic sludge reactor was established, aiming to remove tribromophenol (TBP) efficiently and alleviate the magnetite dissolution.

RESULTS: Under microaerobic condition (DO, 0.6 mg L^{-1}), the efficiencies of removal of COD, TBP, and methane production in Fe_3O_4/AS system were, respectively, 14.3%, 16.7%, and 27.2% higher than those of the AS system. Correspondingly, the dehydrogenase (DHA) and Coenzyme F_{420} (CoF_{420}) activities were separately 1.38-fold and 1.41-fold enhanced. The enhanced biodegradation was via potential DIET, as visualized by microbes and magnetite particles, *Geobacter* enrichment and *c*-type cytochromes (*c*-Cyts) increasement. Notably, magnetite majorly maintained an intact structure. In effluents, the Fe(II) concentrations were maintained at less than 10 mg L^{-1} under microaerobic conditions, far less than the amount under anaerobic conditions, which might be due to the *in situ* redox cycle of Fe(III)-reduction and Fe(II)-reoxidation.

CONCLUSION: Significant enhancements in removal efficiencies of TBP and COD and methanogenesis were demonstrated in a Fe_3O_4/AS system, indicating intrinsically faster electron transfer via electrical conduction. Putative oxidation of dissolved Fe(II) *in situ* occurred in microaerobic environments, which could protect magnetite in the system without extra magnetite supplementation.
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Supporting information may be found in the online version of this article.

Keywords: direct interspecies electron transfer; methanogenesis; microaerobic; submicron magnetite; tribromophenol wastewater

INTRODUCTION

Tribromophenol (TBP) is widely used as a synthetic intermediate of brominated flame retardants (BFRs), pesticide, and the replacement of pentachlorophenol in wood preservation. TBP has developmental neurotoxicity, embryotoxicity, and fetotoxicity effects, and this compound has been detected in various environmental matrices throughout the world, including marine and fresh waters, sediments, soil, sewage sludge, and organisms.[1–5] Notably, wastewater containing TBP has attracted considerable concern due to long-term residual and bioaccumulative properties, and the development of efficient and cost-effective techniques to treat TBP wastewater is needed.

To date, limited physicochemical methods (phototransformation, chemical catalysis, etc.) have been reported to remove TBP, while biodegradation is the eco-friendly and economical approach for removal of TBP.[6] However, TBP is resistant to aerobic biodegradation due to the electron-withdrawing properties of bromine. But anaerobic dehalogenation via dehalorespiration using halogenated compounds as terminal electron acceptors occurs more easily.[7] Previous studies have reported that both activated sludge (AS) and pure strains exhibit anaerobic biodegradability of TBP.[8,9] Lei et al. isolated *Bacillus* sp. GZT isolate, which was capable of debrominating and mineralizing TBP, inferring that anaerobic reductive debromination could act as the major degradation pathway in the process of mineralizing TBP.[10] However, the electron transfer rate is relatively low under anaerobic conditions, which limits the rate of the dehalogenation process. To solve this problem, different mediators have been applied in biodegradation processes as electron shuttles to accelerate electron transfer, such as vitamin B_{12} (VB_{12}) and anthraquinone-2,6-disulfonate (AQDS).[11,12] Recently, it has been documented that nanoparticle magnetite (Fe_3O_4) and other conductive minerals can promote direct interspecies electron transfer (DIET) in a range of metabolic processes, including a syntrophic metabolism between dehalogenating and non-dehalogenating bacteria under anaerobic

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conditions.[13–15] Moreover, it has been well demonstrated that magnetite supplementation has beneficial effects on methanogenesis and process stability by enhanced DIET.[16]

Nonetheless, it has been reported that the added magnetite in AS systems could be reduced by dissimilatory iron reducing bacteria (DIRB) under anaerobic conditions.[17] Releasing biogenic dissolved Fe(II) from magnetite (i.e. Fe(III)-reduction process), may give rise to the undesirable loss and non-sustainability of magnetite in wastewater treatment processes. To fill such a knowledge gap, the application of a microaerobic process may alleviate the dissolution of magnetite by DIRB and conduct the potential Fe(II)-reoxidation in the magnetite-supplemented AS system. Besides, the microaerobic condition offers apparent merits over the conventional anaerobic process due to the high-efficiency of hydrolysis, acidification and complete degradation of the toxic contaminants simultaneously.[18–20]

Hence, in order to promote the magnetite-mediated DIET as well as restore the magnetite *in situ,* this study proposed that submicron magnetite particles supplement the AS system at a microaerobic dissolved oxygen (DO) level (0.3–1.0 mg L^{-1}) to set up a Fe_3O_4/AS system to treat TBP wastewater. The objectives of this study were[1] to elucidate the effects of submicron magnetite particles on TBP removal and methanogenesis under microaerobic conditions[2]; to investigate whether or not the supplemented magnetite is maintained in continuous culture under microaerobic conditions;[3] and to explore the underlying mechanisms in the biodegradation process. This study can provide new possibilities for high-rate TBP removal and biomethanation and help to cope with the dissolution of magnetite in Fe_3O_4/AS systems.

MATERIALS AND METHODS

TBP and synthetic wastewater

TBP (98% purity) was purchased from Aladdin Reagent (Shanghai, China). A stock solution of TBP (5000 mg L^{-1}) was prepared in ethanol (99% purity) and stored at 4 °C in the dark prior to use. The synthetic mineral medium was prepared in deionized water and contained the following components (mg L^{-1}): NH_4Cl (500), KH_2PO_4 (500), K_2HPO_4 (600), $MgCl{\cdot}6H_2O$ (200), $CaCl_2$ (50), $(NH_4)_2SO_4$ (100), and 1 mL L^{-1} of trace elements solutions, which contained H_3BO_3 (100), $CuSO_4{\cdot}5H_2O$ (58), $ZnSO_4{\cdot}7H_2O$ (100), $NiSO_4{\cdot}6H_2O$ (112), $MgSO_4{\cdot}7H_2O$ (5000), $FeSO_4{\cdot}7H_2O$ (750), $MnSO_4{\cdot}H_2O$ (188), and yeast (3000). Glucose as the carbon source was added into the synthetic TBP wastewater to provide an initial COD concentration of 1000 mg L^{-1}. The pH value of the synthetic TBP wastewater was adjusted to 7.0. The physicochemical characteristics of the seed sludge (collected from the secondary clarifier of the Chunliu River municipal wastewater treatment plant in Dalian, China) are shown in Table S1, Supporting information.

Batch experiments

All batch experiments were conducted in the dark at room temperature (20–25 °C) in serum flasks (120 mL) which were supplied with basic mineral medium (100 mL) and inoculated with activated sludge statically to achieve a final mixed liquid suspended solids (MLSS) of 330 mg L^{-1}. Upon preparation, all bottles were sealed with Teflon-faced butyl rubber stoppers and flushed with 99.5% N_2.

Different size scales of magnetite were added into the serum bottles to investigate their effects on biodegradation performance as follows (mixed magnetite particles with AS at a mass ratio of 0.25): AS (control), AS plus nano-Fe_3O_4, AS plus submicron-Fe_3O_4, AS plus micron-Fe_3O_4. After 24 h of inoculation, the bottles were sampled for the determination of remaining COD contents. The submicron magnetite particles (0.3–0.5 μm), micron magnetite particles (30–50 μm), and nanoscale magnetite particles (20 ± 5 nm) were purchased from Aladdin Reagent (Shanghai, China).

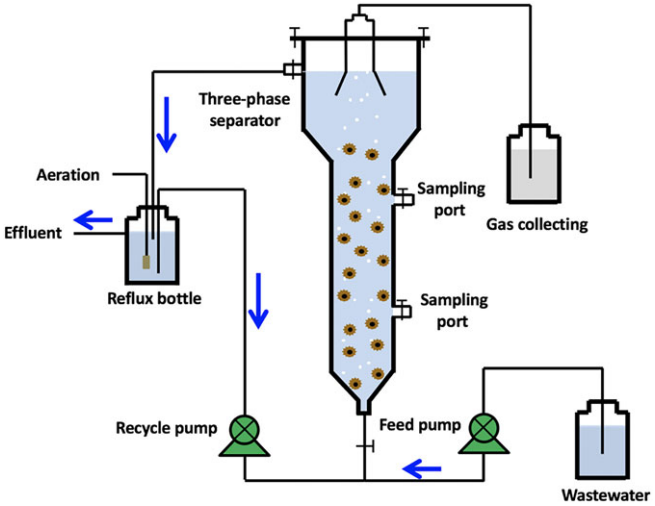


Figure 1. Schematic diagram of reactor configuration.

Additional batch experiments were also conducted to investigate the effects of different concentrations of dissolved Fe(II) on biodegradation. A total of three serum bottles were inoculated as followed: X_c (AS), X_1 (AS plus 10 mg L^{-1} $FeCl_2$), X_2 (AS plus 20 mg L^{-1} $FeCl_2$). After 24 h of inoculation, the remaining COD contents were detected and the methane production was measured by water displacement method. All the tests were performed in duplicate in darkness, in a constant temperature incubator at 35 °C.

The experimental set-up and operation

Two lab-scale microaerobic up-flow sludge reactors were operated with a working volume of 1.35 L, namely R_1 (AS, control) and R_2 (Fe_3O_4/AS), respectively. The initial MLSS in both reactors were around 5000 mg L^{-1}. The reactors were run in batch mode during the first 5 days for start-up and then switched to continuous mode with daily feeding. R_2 was supplemented with submicron magnetite at a mass ratio of 0.25; R_1 was not amended (control). They were continuously fed with synthetic wastewater containing TBP at a controlled flow rate using a peristaltic pump with 50% volume exchange ratio and 1 cycle per day (HRT, 24 h). The influent was purged with 99.5% of N_2 to maintain DO below 0.05 mg L^{-1} and reflux bottles equipped with aeration devices were used to maintain the microaerobic DO condition. Both reactors were maintained at pH ~7 and the temperature was kept at 35 °C± 2 °C using a water bath throughout the experiment. The schematic diagram of reactor configuration is shown in Fig. 1.

The operation of reactors was divided into three periods according to the operating conditions. The time period and operating conditions for each period are tabulated in Table 1. During the operational period (Day 1–65), reactors were acclimated with the increase of COD (1000–2000 mg L^{-1}) and TBP (5–20 mg L^{-1}) at 0.6 mg L^{-1} DO concentration. Anaerobic, microaerobic and aerobic conditions were introduced to evaluate the effect on TBP and COD removal and methanogenesis by different DO concentrations

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Table 1. Operational conditions, test indexes in both reactors during operation

Operational time	DO condition (mg L^{-1})	Influent concentration(mg L^{-1})	Test indices
			Removal efficiencies of COD, TBP
Day1-35		TBP = 5, COD = 1000	Sludge volume index
Day35-55	0.6	TBP = 10, COD = 1500	Fe(II) concentration
Day55-65		TBP = 20, COD = 2000	Sludge size distribution
	1.51		Removal efficiencies of COD, TBP
Day65-70	0.68	TBP = 20	Methanogenesis performance analysis
	0.11	COD = 2000	Fe(II) concentration
			Microbial community analysis
Day70-75	0.6	TBP = 20	Coenzyme F_{420}
		COD = 2000	*c*-type cytochromes
			Dehydrogenase activity
			Iron species analysis

during Day 65–70. The underlying mechanisms including microbiological characteristics analysis and iron species analysis were then investigated.

Analytical methods

COD, MLSS, sludge volume index (SVI) were all determined according to Standard Methods.[21] DO and pH were measured using a digital DO meter (YSI, Model 55, Yellow Springs, Ohio, USA) and a digital pH meter (PHS-25, Leici Company, Shanghai, China). TBP concentrations of influents and effluents were determined by high-performance liquid chromatography (HPLC) (Shimadzu LC-20AT, Kyoto, Japan) equipped with a DAD detector set at 286 nm as described in reference 22. Coenzyme F_{420} was tested by the classic procedure of spectrophotometric study.[23] A sample of 2 mL from the reactor was first disintegrated and suspended in a Bead Bug (D 1030, Benchmark, Sayreville, New Jersey, USA) at 500 oscillations $min^{-}1$ for 100 s, then centrifuged at 10 000 rpm and 4°C for 15 min to remove the waste debris. The suspensions were measured using a fluorescence/luminescence spectrometer (LS 55, Perkin Elmer, Akron, Ohio, USA) under the synchronous scan regime. The difference between excitation and emission wavelengths was 20 nm. The excitation wavelength was changed from 360 nm to 480 nm and the emission wavelength was synchronously changed from 380 nm to 500 nm. Fluorescence at 420 nm was used in the experiments. Particle size of granules was tested with a laser particle analyzer (Mastersizer 2000, Malvern, Royston, England). The association of magnetite particles and bacterial cells on Day 65 was observed and photographed by a scanning electron microscope (SEM) (NOVA NanoSEM 450, FEI, Oregon, USA). The total dehydrogenase (DHA) activity was measured as described by Feng *et al.*[24] A ferrozine colorimetric method for Fe(II) determination was performed using a UV–visible spectrophotometer (V-560, JASCO, Tokyo, Japan) at 560 nm. X-ray diffraction (XRD) analysis was performed using a XRD-7000 diffractometer (Shimadzu) with a Cu Kα radiation source. The extracellular polymeric substances (EPS) were extracted from the samples using a heat technique.[25] The total heme *c* contents and the heme *c* contents in EPS were measured by the methods recorded by Berry *et al.*[26] The cytochrome peroxidase reaction was conducted on the basis of the methods described by Yin *et al.* and the transmission electron microscope (TEM) observation was performed according to the methods described by Duan *et al.* using a Tecnai F30 TEM (FEI).[27,28] Biogas production was measured volumetrically using a water displacement method at room temperature (25–30°C).The results were analyzed and compared using one-way ANOVA with TUKEY's test at the significance level of 0.05.

16S rRNA amplicon sequencing and data processing

At the end of the microcosm incubation, sludge samples were collected and stored at −80 °C until further analyses. Genomic DNA was extracted using the FastDNA Spin Kit for Soil (MP Biomedical, Santa Ana, California, USA) following the protocols. 16S rRNA genes were amplified using the bacterial universal primers (515F and 806R) with specific barcode. All PCR reactions were carried out in 30 µL reactions with 15 µL of Phusion® High-Fidelith PCR Master Mix (New England Biolabs, Ipswich, Massachusetts, USA); 0.2 µmol L^{-1} forward and reverse primers, and ~10 ng template DNA. Thermal cycling consisted of initial denaturation at 98 °C for 1 min, followed by 30 cycles of denaturation at 98 °C for 10 s, annealing at 50 °C for 30 s, and elongation at 72 °C for 30 s. Finally, 72 °C for 5 min. PCR products were quantified, mixed, purified and sequenced on an Illumina HiSeq platform at Novogene (Beijing, China).

Paired-end reads from the DNA products were merged by using FLASH.[29] Paired-end reads were assigned to each sample according to the unique barcode. Sequences were analyzed using QIIME (Quantitative Insights into Microbial Ecology).[30] Ambiguous N, low-quality tags, chimers were filtered to generate high quality tags with default parameters. Sequences with similarity higher than 97% were clustered into an Operational Taxonomic Unit (OTU). Representative sequences were picked for each OTU and annotated using the RDP classifier.[31] Perl scripts were used to analyze the α-diversity indices. In order to compute α-diversity, we rarified the OTU table according to the sample with the least sequences with the script in QIIME. The community richness and α-diversity were estimated by Observed species, Chao1, Shannon, Simpson and ACE. Rarefaction curves were generated to compare the level of bacterial OTU diversity.

RESULTS AND DISCUSSION

Tribromophenol removal performance

The COD removal efficiencies of the Fe_3O_4/AS system with different sizes of magnetite particles are shown in Figure S1, Supporting information. The results showed that the AS + submicron-Fe_3O_4 system exhibited excellent COD removal efficiency (84.5%), compared with AS + micron-Fe_3O_4 (80.2%) and AS + nano-Fe_3O_4 (78.3%). The submicron size had a much larger specific surface

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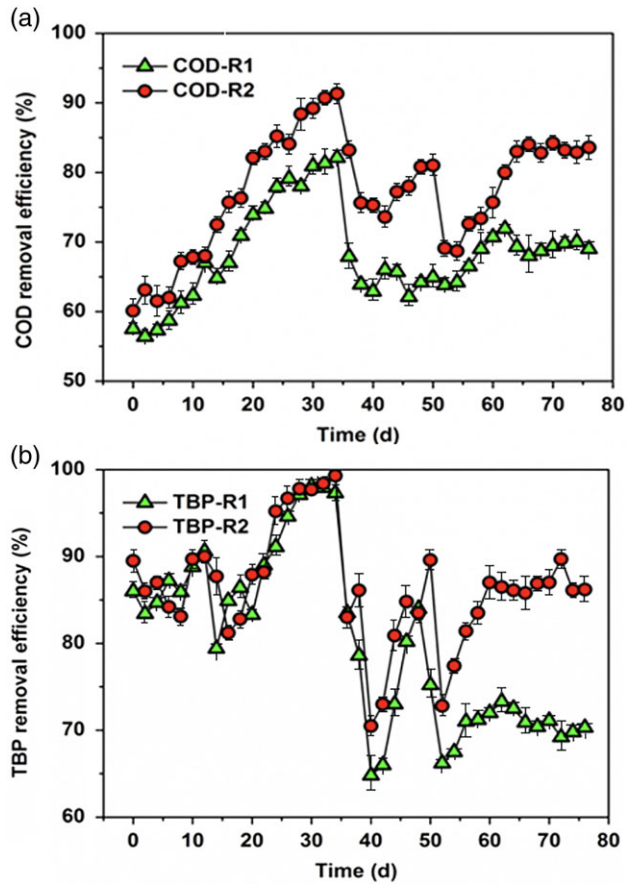


Figure 2. Time course of COD removal efficiency (a) and TBP removal efficiency (b) in both reactors.

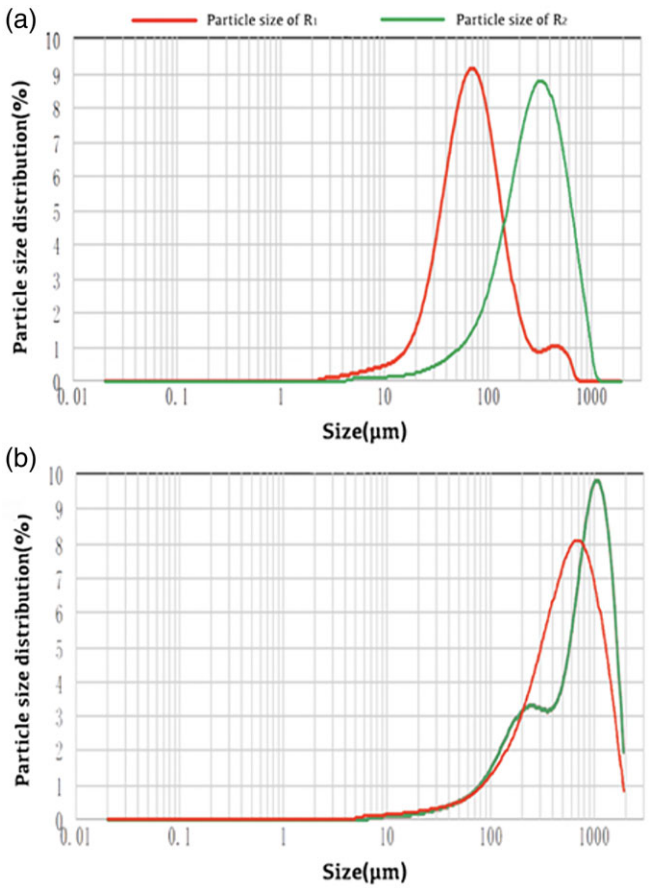


Figure 3. Sludge size distribution in both reactors during operation on day 24 (a) and on Day 60 (b).

area in comparison with the micron size, which could potentially enhance electrons transfer with microbes. Ma *et al.* reported that the tiny particle structure of nanoscale material might penetrate the microbial cell membrane, lead to inactivation and thus reduce the capability of biodegradation.[32] The lower toxicity of submicron magnetite particles could contribute to better bioactivity of the AS/Fe_3O_4 system. Therefore, submicron magnetite particles were supplemented in the following reactor operations.

The removal efficiencies of COD and TBP in R_1 and R_2 are shown in Fig. 2. The initial influent concentrations were COD = 1000 mg L^{-1}, TBP = 5 mg L^{-1}, and the COD removal efficiency of R_2 was 91.3% after 35 days of incubation, which was significantly higher than that of R_1 (9.2%, $P < 0.05$). However, there was no remarkable difference in terms of TBP removal efficiency (>95%) in both reactors. The influent concentration was then altered to COD = 1500, TBP = 10 mg L^{-1} at Day 35. The removal efficiencies of COD and TBP in R_2 were up to 81.0% and 89.6% at Day 50, which were 16.1% and 14.4% higher than those of R_1, respectively. Thus, the mediating effect of the magnetite particles in R_2 had an advantage over R_1 during this period. Subsequently, the influent concentration was enhanced to COD = 2000 mg L^{-1}, TBP = 20 mg L^{-1} at Day 51 and maintained for 25 days. The TBP removal efficiency of R_2 was significantly higher than that of R_1 (average 13.9%, $P < 0.05$). Similarly, the COD removal efficiency of R_2 was significantly higher than that of R_1 (10.2%, $P < 0.05$). The above results showed that the Fe_3O_4/AS system could cope with the higher influent loads of COD and TBP.

Sludge properties were measured to evaluate the stability of AS and Fe_3O_4/AS systems, including granular particle size and SVI. As shown in Fig. 3, the average granular diameters in R_2 were approximately 0.123 and 1.833 mm on Day 24 and Day 60, respectively, which were significantly larger than those of R_1 (0.084 and 1.370 mm). These data revealed that submicron magnetite particles could obviously accelerate the sludge granulation. Time course profiles of SVI on day 1, 30, 45, and 65 are given in Table 2. It was found that the granulation degree of sludge increased with increasing reaction time. The SVI values of R_2 were significantly lower than those of R_1, indicating that the Fe_3O_4/AS system exhibited a specific advantage of settleability over the AS system. By comparison, the Fe_3O_4/AS system could observably shorten the start-up process by promoting the formation of granular sludge.

To investigate the impacts of different DO concentrations on TBP and COD removal, anaerobic, microaerobic and aerobic conditions were employed for reactor operation. As shown in Fig. 4(a), the highest TBP removal efficiency was 89.3% in the Fe_3O_4/AS system at 0.11 mg L^{-1} of DO, significantly higher than that of the AS system (17.4%, $P < 0.05$). The enhanced TBP removal was probably due to the magnetite-mediated DIET with strong dehalogenation performance under anaerobic conditions. However, the COD removal efficiencies decreased to 68.1% and 80.6% in R_1 and R_2, respectively, under anoxic conditions (Fig. 4(b)), owing to the lower hydrolysis acidification efficiency and biodegradable rate. Furthermore, as shown in Fig. 4(d), this process could not be sustainable in long-term operation due to the dissolution of magnetite under anaerobic conditions, consistent with previous reports.[15,16]

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Table 2. Dissolved Fe(II) concentration and SVI value in both reactors during operation

Time (day)	Reactor	Dissolved Fe(II) (mg L^{-1})	SVI (mg L^{-1})
1	R_1	1.21 ± 0.13	150 ± 11
	R_2	2.14 ± 0.11	108 ± 13
30	R_1	0.58 ± 0.03	121 ± 8
	R_2	4.63 ± 0.34	93 ± 5
45	R_1	0.83 ± 0.14	89 ± 6
	R_2	4.75 ± 0.29	72 ± 7
65	R_1	0.66 ± 0.08	67 ± 6
	R_2	6.80 ± 0.20	54 ± 4

Aerobic biodegradation played a crucial role in COD removal (90.1%) at 1.51 mg L^{-1} of DO in R_2 (13.2% higher than that of R_1), whereas anaerobic methanogenesis and TBP removal were inhibited. At 0.68 mg L^{-1} of DO, the removals of TBP and COD in R_2 were 85.2% and 84.4%, respectively, which were significantly higher than those of R_1 by 15.4% and 12.1% ($P < 0.05$). Apart from the enhanced hydrolytic acidification rate and TBP removal, the dissolved Fe(II) was dramatically decreased under microaerobic conditions (Fig. 4(d)). These results suggested that the magnetite was likely maintained under microaerobic condition via oxidizing dissolved Fe(II) to Fe(III), decreasing or even completely eliminating the loss of Fe_3O_4, and thus promoted biodegradation sustainably. The Fe_3O_4/AS system could intensely stimulate biodegradation under various DO conditions compared with the AS system. Furthermore, to maintain magnetite-promoted DIET, microaerobic conditions could provide the possibility of recycling magnetite *in situ* without adding extra magnetite during long-term operation.

Methanogenesis performance

Anaerobic digestion contained steps of hydrolysis, acidification, acetogenesis and methanogenesis. Of which, anaerobic methanogenesis performance could be an indicator for complete mineralization of organic compounds.[33] To probe the effects of submicron magnetite on methanogenesis during the operational period, methanogenesis performance was analyzed under different DO conditions. As shown in Fig. 4(c), methane production of R_2 exceeded that of R_1 by 16.0%, 27.2%, and 21.6% under aerobic, microaerobic, and anaerobic conditions, respectively, reaching 168, 216, and 223 mL day^{-1}, respectively. These data indicated that the Fe_3O_4/AS system had excellent methanogenic activity under different DO conditions. It was found that the methane production in R_2 under microaerobic conditions was remarkably higher than the anaerobic methane production of R_1, which might likely result from the enhanced DIET and the supposed long-term protection of magnetite by microaerobic conditions.

Kida *et al.* demonstrated that CoF_{420} concentration was likely an indicator of the activities of H_2-utilizing methanogens.[34] Moreover, Wang *et al.* showed that CoF_{420} was a H_2 carrier in the methanogenic pathway by utilizing CO_2 and H_2, relying on the

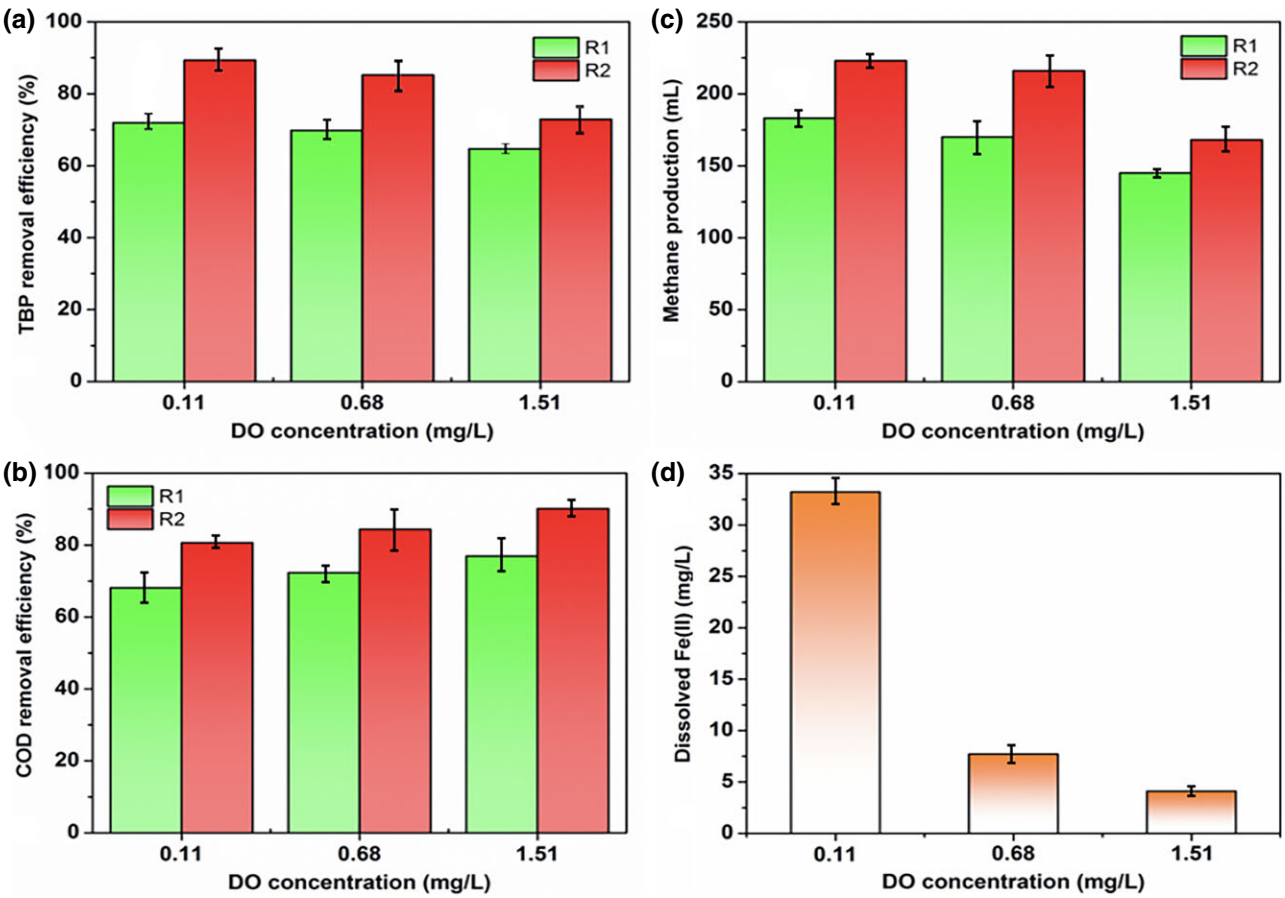


Figure 4. Effect of different DO concentrations on reactors operation: (a) TBP removal efficiencies in reactors. (b) COD removal efficiencies in reactors. (c) Methane production in reactors. (d) Dissolved Fe(II) concentration in R_2 under different DO concentrations.

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Table 3. CoF_{420}, total heme *c* contents, EPS heme *c* contents and DHA activity in both reactors

Reactor	CoF_{420} (μmol g^{-1} MLSS)	Total heme *c* contents (mmol L^{-1} g^{-1} protein)	EPS heme *c* contents (mmol L^{-1} g^{-1} protein)	DHA activity (mg TF/ (L gMLSS h))
R_1	0.126 ± 0.015	1.21 ± 0.23	0.41 ± 0.02	13.47 ± 1.26
R_2	0.178 ± 0.009	1.59 ± 0.16	0.53 ± 0.03	18.59 ± 1.04

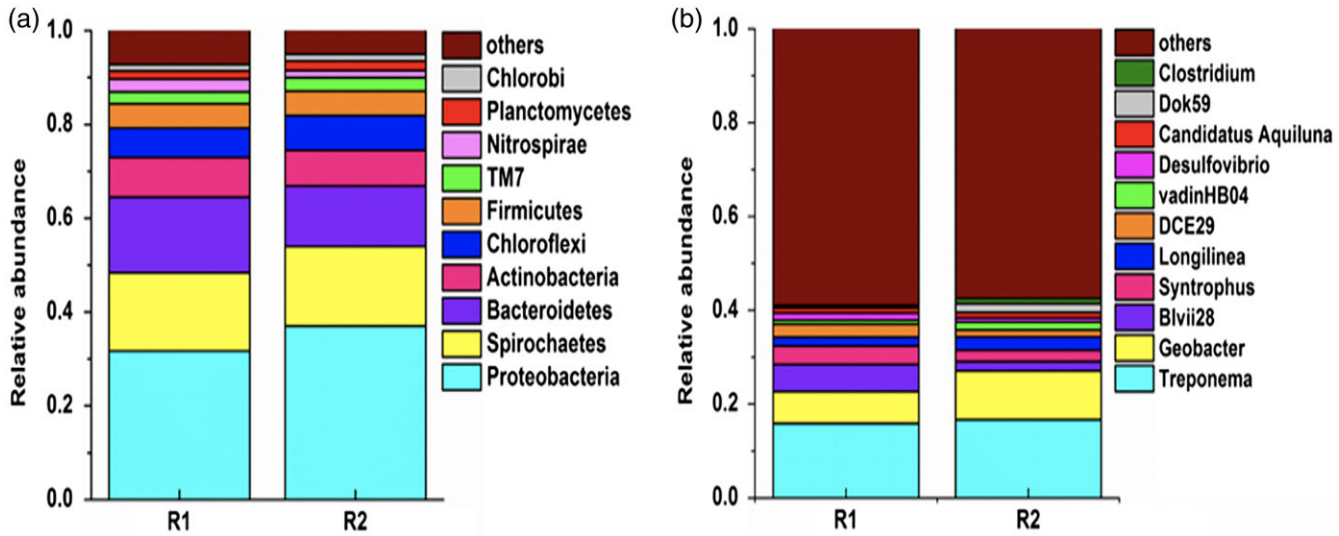


Figure 5. Relative abundance of microbial compositions in both reactors at phylum level (a) and genus level (b).

catalyzation by F_{420}-reducing hydrogenase (Frh).[33] The variation of CoF_{420} concentration in the reactors after operating for 75 days is shown in Table 3. It was found that the CoF_{420} concentration in R_2 reached 0.178 ± 0.009 μmol g^{-1} MLSS, which was 1.41-fold higher than that of R_1 (0.126 ± 0.015 μmol g^{-1} MLSS). Consistent with the better biodegradation performance of R_2, the results implied that supplementation with Fe_3O_4 could greatly stimulate the activities of H_2-utilizing methanogens and increase methane yield, implying that the Fe_3O_4/AS system could perform more steadily and remove organic pollutants efficiently.

Microbial community analysis

A total of 138 818 clean tags (73 548 in R_1 and 65 270 in R_2) were sequenced from two sludge samples collected from both reactors and clustered to 1653 OTUs (1653 in R_1 and 1624 in R_2). *Proteobacteria*, *Spirochaetes*, *Bacteroidetes*, *Actinobacteria*, *Chloroflexi* and *Firmicutes* were the dominant phyla in sludge, accounting for more than 80% of the total bacteria community. The presence of Fe_3O_4 increased *Proteobacteria* and decreased *Bacteroidetes* at the phylum level in R_2 (Fig. 5(a)). The increase of *Proteobacteria* phylum was attributed to the enrichment of *Geobacter* at genus level (Fig. 5(b)). Rarefaction curves in Figure S2 indicate that there was no significant difference in α-diversity between magnetite-amended (R_2) sample and control group (R_1), which was further demonstrated by Shannon, Simpson, Chao1, ACE and goods-coverage indices.

To date, DIET of *Geobacter* co-cultures have been documented in the processes of fumarate reduction, methane production, and nitrate reduction.[13,35,36] The syntrophic metabolism between *Geobacter* and dehalorespiring microorganisms is a tentative mechanism for DIET in the TBP debromination process, which can enhance dehalorespiration by adding excessive electron donors in bioremediation applications.[15] The presence of magnetite particles stimulated the growth of *Geobacter* species of *Proteobacteria* phylum in R_2. In addition, the enriched *Geobacter*, implying potential DIET, could explain the increased methane production and TBP removal efficiency in R_2. Hence, it could be presumed that the enrichment of *Geobacter* species and their syntrophic association with other bacteria (e.g. dehalorespiring bacteria and methanogens) would accomplish reductive debromination and methane production effectively.

c-type cytochromes and dehydrogenase

It has been proposed that *c*-Cyt played important roles in facilitating electron transfer via DIET and was abundant in the periplasm of *Geobacter* sp.[37] For instance, *G. metallireducens* GS-15 had evolved specialized mechanisms in the transfer electrons to *G. sulfurreducens* PCA, the methanogenic archaea *Methanosaeta harundinacea* via OmcS (a multihaem c-Cyt) and nanowires (pili, an extracellular structure formed by protein filaments).[38,39] Hemes, as the electron transfer centers of *c*-Cyts, were detected as the indirect reflection of *c*-Cyts contents. As shown in Table 3, the total heme *c* contents and heme *c* contents in EPS in R_2 were 1.59 ± 0.16 and 0.53 ± 0.03 mmol L^{-1} g^{-1} protein, which were approximately 23.6% and 22.6% higher than those in R_1. The greater amount of heme *c* in R_2 manifested the higher *c*-Cyts level inside the Fe_3O_4/AS system, implying possible stimulation in electron transfer rate.

In order to further investigate the variation of *c*-Cyts in both reactors, the cytochrome reactive histochemical stain 3,3′-diaminobenzidine (DAB) was applied on Day 65. The region of the cytochrome peroxidase stain of cells taken from R_2 was much greater than R_1, consistent with more heme *c* contents, indicating much faster enzymatic reaction engaged in the magnetite-mediated DIET process. Moreover, the microbial cells

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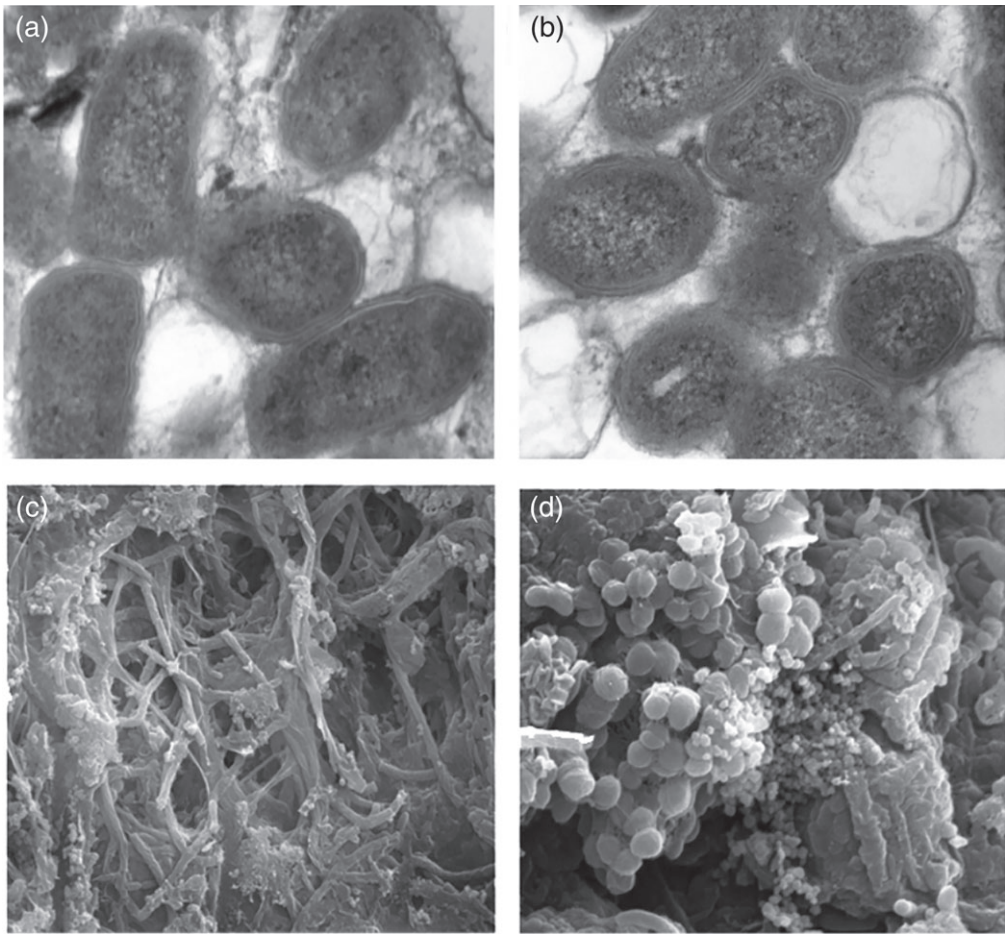


Figure 6. Comparison of the TEM observations in R_1 (a) and R_2 (b). SEM micrographs of microcosms in R_1 (c) and R_2 (d).

in R_2 were more closely cellularly contacted compared with the cell clusters in R_1 and thus made it more convenient for the DIET process (Fig. 6(a) and (b)). Liu *et al.* demonstrated that magnetite nanoparticles could function similarly to *c*-Cyts to facilitate shorter range extracellular electron transfer effectively, compensating for the lack of *c*-Cyts and nanowires in conductivity enhancement.[35] Aulenta *et al.* showed that the magnetite nanoparticles could promote DIET from *G. sulfurreducens* PCA to *Desulfitobacterium* spp. and *Dehalococcoides* spp. in a trichloroethene (TCE) reductive degradation process.[15] Accordingly, we inferred that the submicron magnetite particles, as the electron conduits, had a substantial stimulatory effect on TBP removal and methane production by promoting potential DIET between *Geobacter* and other functional bacteria.

As shown in Table 3, the DHA activity was greatly enhanced in the presence of submicron magnetite particles in R_2 and reached 18.59 ± 1.04 mg TF/ (L gMLSS h) on Day 65 (1.38-fold higher than that of AS system). The microbial activity on organic matter removal and electron transfer could be reflected by the total DHA activity.[40,41] Therefore, the increased DHA contents due to submicron magnetite in R_2 removed the contaminants more effectively and accelerated the electron transfer among cells.

Electrically conductive minerals such as magnetite particles could serve as electron conduits in the proposed DIET mechanism, which required an intimate association between magnetite particles and microbial cells. Sludge samples were taken from both reactors for morphological analysis on Day 65. In comparison with R_1 (Fig. 6(c)), the cells in R_2 appeared tightly contacted and the submicron magnetite crystals appeared as punctuated particles evenly distributed and intimately associated with microbial cells, with EPS attached to the cell surface (Fig. 6(d)). These results indicated that the composite structure of Fe_3O_4/AS could shorten the electron transfer distance between interspecies consortia, potentially creating a network comprised of microorganisms and conductive minerals, thus providing support of a magnetite-mediated DIET mechanism. We propose that the associated structure of magnetite with microbial consortia might increase the abundance of *Geobacter* sp., *c*-Cyts contents, and DHA activity, thus enhancing potential DIET sequentially and providing more energy to syntrophic metabolism and pollutants removal.

Mineral phase analysis

To further explore whether the magnetite particles convert into other forms of iron oxides, the Fe_3O_4/AS complex taken from R_2 was subjected to XRD analysis (Fig. 7). Magnetite and goethite were observed and identified with JCPDS Card No. 11-0614 and JCPDS Card No. 03-0249 respectively, which indicated that the added magnetite was partly transformed into goethite during operation. Goethite, as a kind of semi-conductor, could enhance anaerobic biodegradation and methanogenesis by assisting DIET.[42] Thus, both the added magnetite and the generated goethite in the Fe_3O_4/AS system might have positive effects on DIET.

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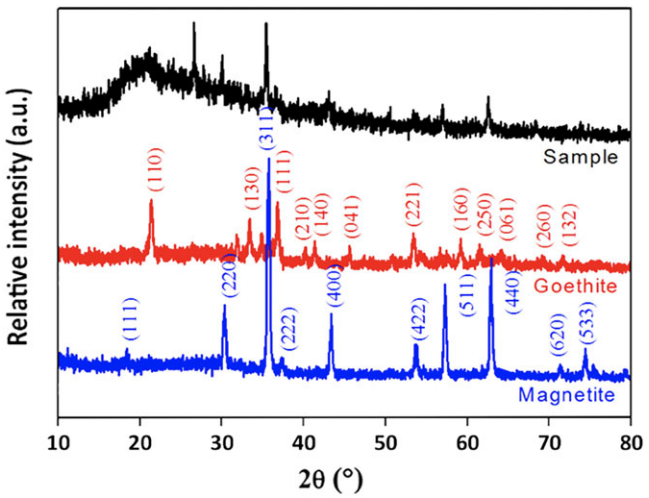


Figure 7. XRD micrographs of Fe_3O_4/AS complex in R_2.

Role of dissolved Fe(II) on TBP biodegradation and Methanogenesis

The dissolution of added magnetite was obtained by periodically measuring the dissolved Fe(II) concentration in the effluents during reactor operation (Table 2). As shown in Fig. 2(d), it was surprising that the microaerobic condition enabled the dissolved Fe(II) to maintain a relatively low concentration (6.8 mg L^{-1} on Day 65), which was far less than the amount under anaerobic conditions (33.2 mg L^{-1} on Day 65). As mentioned above, DIRB utilized Fe(III) as terminal electron acceptors for their growth under anoxic conditions, which was assumed to result in the increase of dissolved Fe(II) concentration.[16] Hence, the microaerobic condition largely mitigated the consumption of magnetite by DIRB and accelerated the biodegradation of organic contaminants simultaneously, which might prolong the function time of the added magnetite material in long-term operation.

To further investigate the effects of different concentrations of dissolved Fe(II) on biodegradation, the results of batch experiments are shown in Fig. S3. Compared with the control group X_c (AS), X_1 (AS +10 mg L^{-1} Fe(II)) and X_2 (AS +20 mg L^{-1} Fe(II)) promoted COD removal and methane production to different degrees, indicating that the presence of Fe(II) enhanced the activity of anaerobic microorganisms. The COD removal efficiency and the methane production in X_1 system were 11.7% and 28.1% higher than those of X_c, indicating that the relatively low concentration of dissolved Fe(II) (<10 mg L^{-1}) released in R_2 reactor had a profound influence on the biodegradation and methanogenesis, which was consistent with Wang *et al.*[33]

CONCLUSIONS

It was demonstrated that remarkable enhancements in the removal efficiencies of TBP and COD and methanogenesis through adding submicron magnetite particles in activated sludge reactors. The direct visualization of the association of magnetite with microbes (SEM), the enrichment of *Geobacter* sp. and the increased contents of *c*-Cyts in Fe_3O_4/AS system indicated the potential magnetite-mediated DIET process, which provided intrinsically faster electron transfer via electrical conduction and thus achieved the excellent biodegradation performances. It was found that putative oxidation of dissolved Fe(II) *in situ* occurred in microaerobic environments and protected magnetite in the Fe_3O_4/AS system without extra magnetite supplementation. The findings of this study could provide possibility for the more efficient treatment of TBP wastewater by modified activated sludge reactors.

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Supporting Information

Supporting information may be found in the online version of this article.

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